

Review

AI-induced hyper-learning in humans

Moshe Glickman^{1,2} and Tali Sharot^{1,2,3}

Humans evolved to learn from one another. Today, however, learning opportunities often emerge from interactions with AI systems. Here, we argue that learning from AI systems resembles learning from other humans, but may be faster and more efficient. Such ‘hyper learning’ can occur because AI: (i) provides a high signal-to-noise ratio that facilitates learning, (ii) has greater data processing ability, enabling it to generate persuasive arguments, and (iii) is perceived (in some domains) to have superior knowledge compared to humans. As a result, humans more quickly adopt biases from AI, are often more easily persuaded by AI, and exhibit novel problem-solving strategies after interacting with AI. Greater awareness of AI’s influences is needed to mitigate the potential negative outcomes of human-AI interactions.

Addresses

¹ Affective Brain Lab, Department of Experimental Psychology, University College London, London, UK² Max Planck UCL Centre for Computational Psychiatry and Ageing Research, University College London, London, UK³ Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA, USACorresponding authors: Glickman, Moshe (mosheglickman345@gmail.com); Sharot, Tali (t.sharot@ucl.ac.uk)

Artificial Intelligence (AI) systems are often designed to learn from humans. However, humans were not designed to learn from AI systems. Until recently, most of our learning opportunities came from other humans, either in the form of direct face-to-face interactions or in the form of indirect interactions (via books, art, films, etc.). Yet, as we welcome AI entities into our ecosystem, less learning may occur via human-human interactions and more via human-AI interactions. Thus, given the pervasive scale of human-AI interactions in our lives, understanding the dynamics of learning from AI systems becomes crucial.

Current Opinion in Psychology 2024, **60**:101900

This review comes from a themed issue on Nudges (2025)

Edited by Brian Bauer and N.B. Schmidt

For complete overview about the section, refer [Nudges \(2025\)](#)

Available online 11 September 2024

<https://doi.org/10.1016/j.copsyc.2024.101900>2352-250X/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Here, we argue that human learning (defined as a change in knowledge, beliefs, behavior or skills) from AI systems will likely resemble learning from other humans. However, it may be faster and more efficient, a phenomenon we refer to as ‘hyper-learning’. We provide three documented examples of such hyper-learning: AI induced bias, AI persuasion and AI induced creativity, and explore the potential mechanisms underlying hyper-learning from AI systems. These examples highlight both the positive and negative outcomes of hyper-learning from AI systems. With regard to the latter, we underscore the need for strategies to minimize unwanted AI influence on humans.

When and why do human-AI interactions lead to more biased humans?

While many AI systems generally produce unbiased judgments, there are clear cases of biased algorithms [1,2,3]. Biases in AI systems tend to arise in two cases. The first is when the data used to train an algorithm is more representative of one class than the other, resulting in less accurate predictions for the underrepresented classes [4]. A well-known example is the difficulty of AI facial recognition algorithms in perceiving non-white faces, as they have been trained predominantly on white faces (e.g., Ref. [5]). The second case is when biases in AI systems arise because of existing human biases embedded in the datasets the algorithm was trained on (‘bias in bias out’; [6], see also [7]). For example, an algorithm used by Apple was found to assign greater financial credit to males than to equivalent female clients [8], a hiring algorithm used by Amazon favored male applicants over equivalent female applicants [9], and Google’s search engine offered more images of men than women in response to the gender-neutral query ‘person’, especially in countries with larger gender inequality [10].

While these cases have been well documented [11,12], an even greater issue may have been largely overlooked. Critical decisions are increasingly made jointly by humans and AI systems. These human-AI interactions provide a mechanism by which, not only biased humans generate biased AI systems, but also biased AI systems can alter human decision-making processes, potentially leaving humans more biased than they were before.

For example, Large Language Models (LLMs) such as ChatGPT, Gemini and Claude, learn from available data on the internet, which being generated by humans, contains many inaccuracies and biases, even in cases where the ground truth exists. As a result, these AI systems end up reflecting a host of human biases such as the conjunction fallacy and the bat-and-ball bias, to name a few (see Ref. [13,14]). When humans then interact with these LLMs, by asking questions and receiving advice, they may learn from the models in return. Similar circularity can also be observed in interactions with other AI systems that exhibit bias, such as text-to-image generative AI models (Luccioni et al., 2023) and recommendation algorithms [15].

Consider the case of Stable Diffusion [16], a text-to-image generative algorithm used by millions [17]. When prompted to generate photos of individuals in high-power and high-income professions, such as lawyers or doctors, Stable Diffusion generates images that over-represent white men compared to other demographic groups ([18]; Luccioni et al., 2023). When humans are then exposed to these photos, either by direct interaction with Stable Diffusion or simply by viewing these outputs passively (e.g., in online ads), they become more biased themselves. For instance, we [19] found that after participants were exposed to images of financial managers generated by Stable Diffusion, they were more likely to pick images of white men to be financial managers over images of individuals of another gender and/or ethnicity, relative to before they interacted with Stable Diffusion.

Similarly, [20] showed that individuals assisted by a biased AI system in a medical diagnostic task reproduced the model's bias in their decisions, even when they later worked without AI support. Indeed, the tendency of humans to become more biased after interacting with a biased AI generalizes to many other domains and judgment types, including motion discrimination, emotional perception and a range of social judgments [19].

Are these examples of AI induced bias qualitatively or quantitatively different from traditional human-human learning? The answer appears to be yes. Human-AI interactions seem to lead to greater and faster adoption of bias than human-human interactions due to two factors [19]: First, the signal-to-noise ratio of AI output is, on average, greater than that of humans. Human outputs (that is, human judgments and opinions available to another human to learn from) are inherently noisy [21]. This is true both when output originates from the same person, due to altered emotional and cognitive states of an individual, and when the output originates from different individuals, due to differences in the individuals' experiences, prior beliefs, etc. AI system responses, on the other hand, tend to show less variability

(see for example [22]). The higher signal-to-noise ratio of AI input makes it easier for humans to take up the signal.

Second, in a growing number of domains, humans perceive AI systems as more capable and accurate than other humans [23,24] and thus are more open to learn from them. Indeed, by having participants interact online with either an AI partner or a human partner and labeling the partners accurately in some conditions while mixing up the labels in others (i.e., participants believe they are interacting with an AI when interacting with a human and vice versa), we were able to dissociate the effect of participants' perception of their partner from the input they were receiving [19]. The results showed that both the stronger signal-to-noise ratio in AI input, and the perception of AI as more accurate, made learning from AI systems more efficient compared to learning from other humans.

When and why are AI systems more persuasive than humans?

Persuasion has traditionally been considered a human endeavor, requiring sophisticated use of language, emotion, and reasoning to influence others' beliefs. Yet, studies have shown that LLMs, such as ChatGPT, exhibit similar or even greater persuasion effectiveness than humans in various topics, including highly polarized issues like abortion and gun control [25,26]. For example, in one study, participants were 81.7% more likely to be convinced by ChatGPT-4 than by a human debater [26].

There are several reasons why AI may be more persuasive than the average human. First, AI is able to process and analyze vast amounts of data, creating more informed and logical arguments than those created by humans [25]. Second, AI systems are more likely to personalize arguments based on the individual's preferences and personality compared to humans [26,27], rendering them more effective. Third, humans may perceive AI systems as more knowledgeable than humans and as lacking ulterior motives. Interestingly, human perception of AI motives is highly malleable. In a recent study [28], participants who were primed with statements indicating that an LLM was either caring, manipulative or neutral, perceived it as such when interacting with it. Moreover, when participants were told an LLM was manipulative, they interacted with it in a negative manner which led to more negative reactions from the AI system, confirming participants' negative expectations. Conversely, when participants were told the LLM had caring motives, they were more likely to generate positive statements when interacting with it, which led it to respond in a similar manner, which in turn reinforced their belief in the AI's kindness and goodwill. This dynamic indicates a powerful cycle where initial perceptions can significantly affect the

outcome of human-AI interactions, with the influence of an AI system being larger when people believe the system was designed to have caring motives.

Another factor that impacts the extent to which an AI system influences humans is their perception of the AI's confidence in its knowledge. It has been shown that people overestimated the confidence of AI agents, even when the AI performance is identical to that of humans [29], which makes them more likely to accept the AI agent's advice over that of other humans.

While labeling content as AI-generated can increase its influence, this is not always the case. For example, pro-vaccination messages generated by AI were found to be more convincing when people believed them to be human-generated [30]. Similarly, messages of empathy generated by an AI were much better received when individuals believed they were human-generated [31]. More research is needed to determine the conditions under which AI-generated content is more persuasive than human-generated content, and when it is less so.

We predict that as LLMs (and other AI systems) continue to evolve, their persuasive power will increase further. This may present a problem in cases where humans employ these systems for political gain or to spread misinformation.

AI-human interactions can lead to more creative humans

On the positive side, AI also has the potential to significantly enhance human performance both directly by providing implementable advice or solutions [32,33] or indirectly, by enabling humans to learn novel approaches and strategies previously unexplored or unrecognized [34].

A compelling example of the latter was observed in human Go players who interacted with AlphaGo, an AI algorithm trained to play the ancient game. By analyzing millions of game moves, [34] found that human Go players began incorporating novel moves into their gameplay after interacting with AlphaGo, leading to better overall performance. Despite the fact that Go has been played for centuries, the players seemed to have learned completely novel strategies developed by AlphaGo — previously undiscovered by humans.

Similarly, a recent study [35] demonstrated that learning from algorithms can enhance human problem-solving compared to learning from other humans. In this study, participants completed a network navigation task. Each node in the network was connected to other nodes, which were associated with rewards and losses.

Participants had to find the optimal sequence of moves that maximizes their payoff. Before attempting the task on their own, participants observed another participant's performance, who was either a human or an algorithm. The findings revealed that participants' performance improved more after observing solutions provided by algorithms due to exposure to novel strategies not commonly used by humans. For example, algorithms frequently selected routes that initially reduced the reward but eventually resulted in higher overall ones. Participants were thus able to observe and adopt unconventional strategies that are not typically generated by humans. However, the study also identified a significant challenge: the participants under-adopted innovative algorithm-derived strategies due to inherent human biases that favor familiar solutions.

Conclusions

In conclusion, human learning from AI may be faster and more efficient than from other humans due to greater signal-to-noise ratio in the input of the AI, its ability to process large amounts of data and clearly compose arguments, and (in some domains) people's belief that AI systems are more knowledgeable than humans. Evidence of such hyper-learning is beginning to emerge in the literature. Studies show that humans learn to adopt biases faster from an AI system than from a human [10,20,19], that they are more likely to be persuaded by AI on different issues [26], and that they are quick to exhibit novel and creative problem-solving strategies after interacting with an AI system [34,35]. AI-induced hyper-learning may prove to be beneficial in a range of other domains [36,37,38]. For example, AI-powered platforms, such as chatbots, can tailor cognitive behavioral therapy (CBT) exercises based on real-time feedback, potentially facilitating skill acquisition that improves mental health. Indeed, chatbots like Wysa have been reported to effectively deliver CBT and mindfulness exercises, leading to reduced symptoms of depression and anxiety [39]. Further research, however, is needed to validate these methods and explore their long-term implications for mental health treatment.

Human hyper-learning from AI can be beneficial when the outcome is an improvement in cognitive abilities [22,34,35]. However, alongside this promise are concerns that interactions with AI will result in stronger human bias [10,20,19], including social biases, and unwanted persuasion by AI, for example in political domains. How can we mitigate these threats? One approach is to focus on improving AI systems themselves, for example conducting algorithmic audits, identifying biased predictions and establishing guidelines and best practices for designing and deploying AI systems [15]. However, it is also essential to increase

human awareness of these potential problems and provide educational programs to increase AI literacy [40]. Indeed, our own research shows that humans are often unaware of AI's influence on them [19]. Therefore, it is important to educate the public about the potential risks and benefits of AI. This is particularly true for children who have more malleable knowledge structures, and thus may be more susceptible to the influence of AI systems than adults [41].

Author contributions

Moshe Glickman: Writing – original draft preparation, Writing – review & editing **Tali Sharot:** Writing – original draft preparation, Writing – review & editing, Supervision

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

T.S. is funded by a Wellcome Trust Senior Research Fellowship 214268/Z/18/Z.

References

References of particular interest have been highlighted as:

- * of special interest
- ** of outstanding interest

1. Zou J, Schiebinger L: **Design AI so that it's fair**. *Nature* 2018, **559**:324–326.
2. Lloyd K: **Bias amplification in artificial intelligence systems**. 2018. arXiv preprint arXiv:1809.07842.
3. Hall M, van der Maaten L, Gustafson L, Jones M, Adcock A: **A systematic study of bias amplification**. 2022. arXiv preprint arXiv:2201.11706.
4. Geirhos R, Rubisch P, Michaelis C, Bethge M, Wichmann FA, Brendel W: **ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness**. arXiv preprint arXiv:1811.12231 2018.
5. Buolamwini J, Gebru T: **Gender shades: intersectional accuracy disparities in commercial gender classification**. In *Conference on fairness, accountability and transparency*. PMLR; 2018:77–91. January.
6. Mayson SG: **Bias in, bias out**. *Yale Law J* 2019;128.
7. Peterson JC, Uddenberg S, Griffiths TL, Todorov A, Suchow JW: **Deep models of superficial face judgments**. *Proc Natl Acad Sci USA* 2022, **119**, e2115228119.
8. Nasiripour S, Natarajan S: **Apple Co-founder says goldman's Apple card algorithm discriminates**. Bloomberg; 2019. Retrieved from: <https://www.bloomberg.com/news/articles/2019-11-10/apple-co-founder-says-goldman-s-apple-card-algo-discriminates>.
9. Dastin J: **Amazon scraps secret AI recruiting tool that showed bias against women**. In *Ethics of data and analytics*. Auerbach Publications; 2018:296–299.
10. Vlasceanu M, Amodio DM: **Propagation of societal gender inequality by internet search algorithms**. *Proc Natl Acad Sci USA* 2022, **119**, e2204529119.
11. Caliskan A, Bryson JJ, Narayanan A: **Semantics derived automatically from language corpora contain human-like biases**. *Science* 2017, **356**:183–186.
12. Bolukbasi T, Chang KW, Zou JY, Saligrama V, Kalai AT: **Man is to computer programmer as woman is to homemaker? debiasing word embeddings**. *Advances in neural information processing systems* 2016, **29**.
13. Binz M, Schulz E: **Using cognitive psychology to understand GPT-3**. *Proc Natl Acad Sci USA* 2023, **120**, e2218523120.
14. Yax N, Anlló H, Palminteri S: **Studying and improving reasoning in humans and machines**. 2023. arXiv preprint arXiv:2309.12485.
15. Morewedge CK, Mullainathan S, Naushan HF, Sunstein CR, Kleinberg J, Raghavan M, Ludwig JO: **Human bias in algorithm design**. *Nat Human Behav* 2023:1–3.
16. Rombach R, Blattmann A, Lorenz D, Esser P, Ommer B: **High-resolution image synthesis with latent diffusion models**. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*; 2022:10684–10695.
17. Stability AI. (n.d.). <https://stability.ai/about>.
18. Bianchi F, Kalluri P, Durmus E, Ladhak F, Cheng M, Nozza D, Caliskan A: **Easily accessible text-to-image generation amplifies demographic stereotypes at large scale**. In *Proceedings of the 2023 ACM conference on fairness, accountability, and transparency*; 2023:1493–1504.
19. Glickman M, Sharot T: **How human-AI feedback loops alter human perceptual, emotional and social judgements**. *Nat Human Behav* 2024 (in press).
20. Vicente L, Matute H: **Humans inherit artificial intelligence biases**. *Sci Rep* 2023, **13**, 15737.
21. Kahneman D, Sibony O, Sunstein CR: **Noise: a flaw in human judgment**. Little, Brown; 2021.
22. Chen Y, Liu TX, Shan Y, Zhong S: **The emergence of economic rationality of GPT**. *Proc Natl Acad Sci USA* 2023, **120**, e2316205120.
23. Bogert E, Schechter A, Watson RT: **Humans rely more on algorithms than social influence as a task becomes more difficult**. *Sci Rep* 2021, **11**:1–9.
24. Logg JM, Minson JA, Moore DA: **Algorithm appreciation: people prefer algorithmic to human judgment**. *Organ Behav Hum Decis Process* 2019, **151**:90–103.
25. Bai H, Voelkel J, Eichstaedt J, Willer R: **Artificial intelligence can persuade humans on political issues**. 2023.
26. Salvi F, Ribeiro MH, Gallotti R, West R: **On the conversational persuasiveness of large language models: a randomized controlled trial**. 2024. arXiv preprint arXiv:2403.14380.
27. Simchon A, Edwards M, Lewandowsky S: **The persuasive effects of political microtargeting in the age of generative artificial intelligence**. *PNAS nexus* 2024, **3**, pgae035.
28. Pataranutaporn P, Liu R, Finn E, Maes P: **Influencing human-AI interaction by priming beliefs about AI can increase perceived trustworthiness, empathy and effectiveness**. *Nat Mach Intell* 2023, **5**:1076–1086.
29. Colombatto C, Fleming S: **Illusions of confidence in artificial systems**. 2023.
30. Karinshak E, Liu SX, Park JS, Hancock JT: **Working with AI to persuade: examining a large language model's ability to generate pro-vaccination messages**. *Proceed ACM on Human-Computer Interaction* 2023, **7**:1–29.
31. Perry A: **AI will never convey the essence of human empathy**. *Nat Human Behav* 2023, **7**:1808–1809.
32. Senior AW, Evans R, Jumper J, Kirkpatrick J, Sifre L, Green T, Hassabis D: **Improved protein structure prediction using potentials from deep learning**. *Nature* 2020, **577**:706–710.

33. Fawzi A, Balog M, Huang A, Hubert T, Romera-Paredes B, Barekattain M, Kohli P: **Discovering faster matrix multiplication algorithms with reinforcement learning**. *Nature* 2022, **610**: 47–53.
34. Shin M, Kim J, van Opheusden B, Griffiths TL: **Superhuman artificial intelligence can improve human decision-making by increasing novelty**. *Proc Natl Acad Sci USA* 2023, **120**, e2214840120.
35. Brinkmann L, Gezerli D, Kleist KV, Müller TF, Rahwan I, Pescetelli N: **Hybrid social learning in human-algorithm cultural transmission**. *Philosophical Transac Royal Soci A* 2022, **380**, 20200426.
36. Koutsouleris N, Hauser TU, Skvortsova V, De Choudhury M: **From promise to practice: towards the realisation of AI-informed mental health care**. *The Lancet Digital Health* 2022, **4**:e829–e840.
37. D'Alfonso S: **AI in mental health**. *Curren opin psychol* 2020, **36**: 112–117.
38. Ettman CK, Galea S: **The potential influence of AI on population mental health**. *JMIR Mental Health* 2023, **10**, e49936.
39. Inkster B, Sarda S, Subramanian V: **An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: real-world data evaluation mixed-methods study**. *JMIR mHealth and uHealth* 2018, **6**, e12106.
40. Farahany NA: *The battle for your brain: defending the right to think freely in the age of neurotechnology*. St. Martin's Press; 2023.
41. Kidd C, Birhane A: **How AI can distort human beliefs**. *Science* 2023, **380**:1222–1223.

Further information on references of particular interest

15. Algorithms often fail to reflect users' true preferences because they rely on revealed preferences (user behavior) to make predictions. ^{*} Revealed preferences can be an incomplete or misleading measure of users' actual preferences due to psychological biases and bounded rationality. To design algorithms that better serve users and improve welfare, designers should also consider factors such as users' normative preferences, expanding the range of observed behaviors, and using methods like preference elicitation.
19. This paper demonstrates that human-AI interactions can create a ^{**} feedback loop that amplifies even small biases, leading to increased biases in human judgment across various domains such as perception, emotion and social judgment. The amplification effect is greater in human-AI interactions compared to human-human interactions, due to both the AI's tendency to amplify biases and human perception of AI. People are often unaware of the extent to which AI has influenced their judgments, making them more susceptible to its influence.
26. This paper shows that AI-driven personalized persuasion is ^{**} significantly more effective than human persuasion. When participants debated with a large language model (GPT-4) that had access to their personal information, they were 81.7% more likely to agree with their opponent's position compared to when they debated with other humans.
34. The emergence of AlphaGo, a superhuman AI in the game of Go, ^{**} has significantly impacted the decision-making of professional human Go players. After being exposed to the moves and strategies of AlphaGo, human players' decision quality improved and novel moves occurred more frequently.